

Chapter 10.A

Covalent Bonding

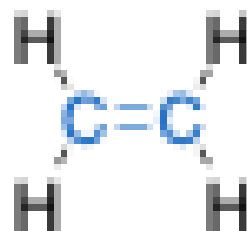
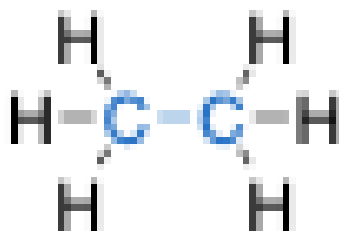
Chemical Principles I

CHEM 1211K

1. Describe and apply essential ideas about the energetics of covalent bonding.
2. Define and apply electronegativity and its periodic trends.
3. Apply bond enthalpies to calculate enthalpies of reaction.
4. Draw valid and plausible Lewis structures for molecules that follow the octet rule.
5. Determine formal charges in Lewis structures by counting electrons.
6. Apply typical exceptions to the octet rule when drawing Lewis structures.
7. Define and apply the concept of resonance to molecules containing delocalized valence electrons; draw valid and plausible resonance structures.
8. Rank the relative contributions of inequivalent resonance forms to the resonance hybrid.
9. Determine the direction and magnitude of bond polarity using electronegativity values.

10.1 Covalent Bonding

- A covalent bond involves the sharing of valence electrons between two atoms (typically nonmetals). The two bonded atoms share (“co-”) valence shells (“-valent”).
 - Covalently bonded atoms may share two electrons (single bond), four electrons (double bond), or six electrons (triple bond).



10.1 Energetics of Covalent Bonding

- The energetics of covalent bonding involve a balance between electron-nucleus attraction and electron-electron repulsion.
 - The internuclear distance at the minimum of energy is called the **bond length r_e** .
 - The depth of the energy well is called the **bond dissociation energy (D_e or BDE)**.

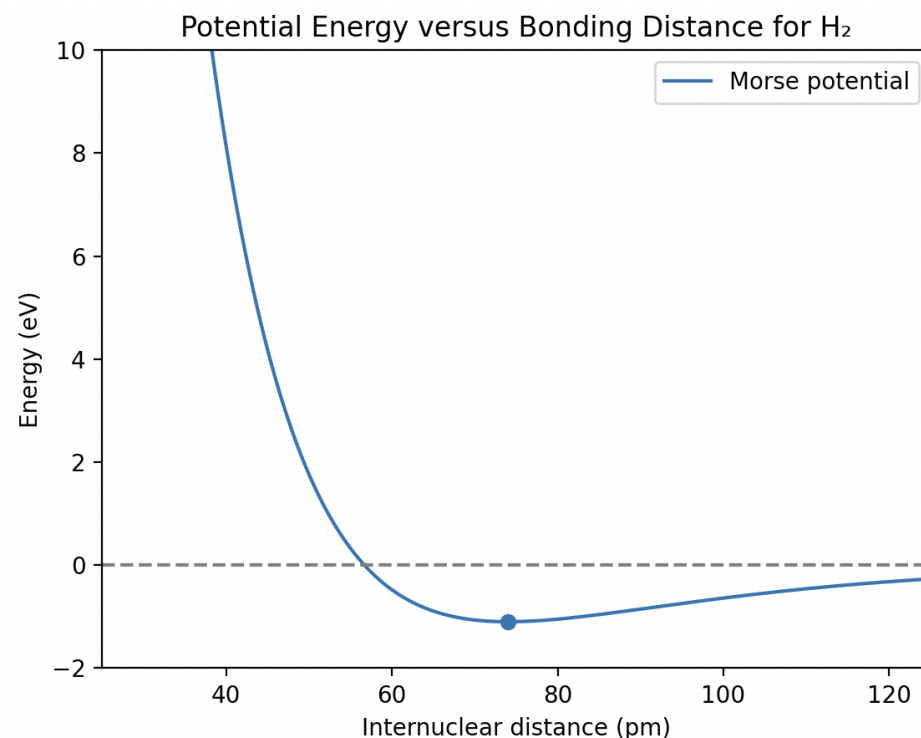
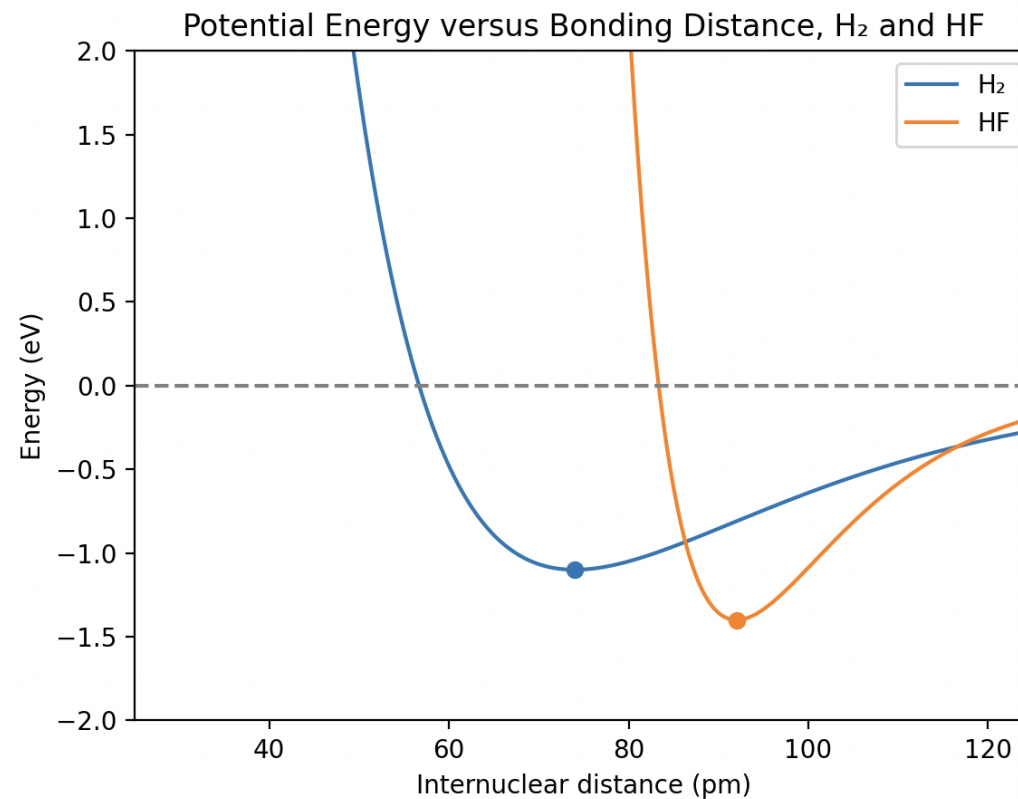


Figure. Energy as a function of the distance between the nuclei in an H₂ molecule.

The graph below shows potential energy as a function of internuclear distance for H_2 and HF . Determine the stronger of the two bonds and the shorter of the two bonds from the graph.

- A. The stronger bond is H–H and the shorter bond is H–H.
- B. The stronger bond is H–F and the shorter bond is H–H.
- C. The stronger bond is H–H and the shorter bond is H–F.
- D. The stronger bond is H–F and the shorter bond is H–F.



- **Electronegativity χ** is the capacity or ability of an atom within a molecule to attract electrons to itself.

- Deep conceptual correlations with Z_{eff} , ionization energy, and electron affinity
- Different quantitative scales have been developed for electronegativity over the years.
 - Mulliken: the average of ionization energy and electron affinity.

$$\chi = \frac{IE + EA}{2}$$

- [Pauling](#): hydrogen is arbitrarily set to 2.20 and differences in electronegativity between elements A and B are defined by the square root of the difference between the bond dissociation energy of AB and the average bond dissociation energy of AA and BB (all in eV).

$$|\chi_A - \chi_B| = \sqrt{D_e(AB) - \frac{D_e(AA) + D_e(BB)}{2}}$$

10.3 Periodic Trends in Electronegativity

- Electronegativity increases left to right across a period and decreases down a group.
This is the most important periodic trend!
- More electronegative elements have greater electron density in covalent bonds (electrons are *unequally* shared!).
- Coincidentally, they also generally form fewer bonds and wind up on the periphery of Lewis structures.

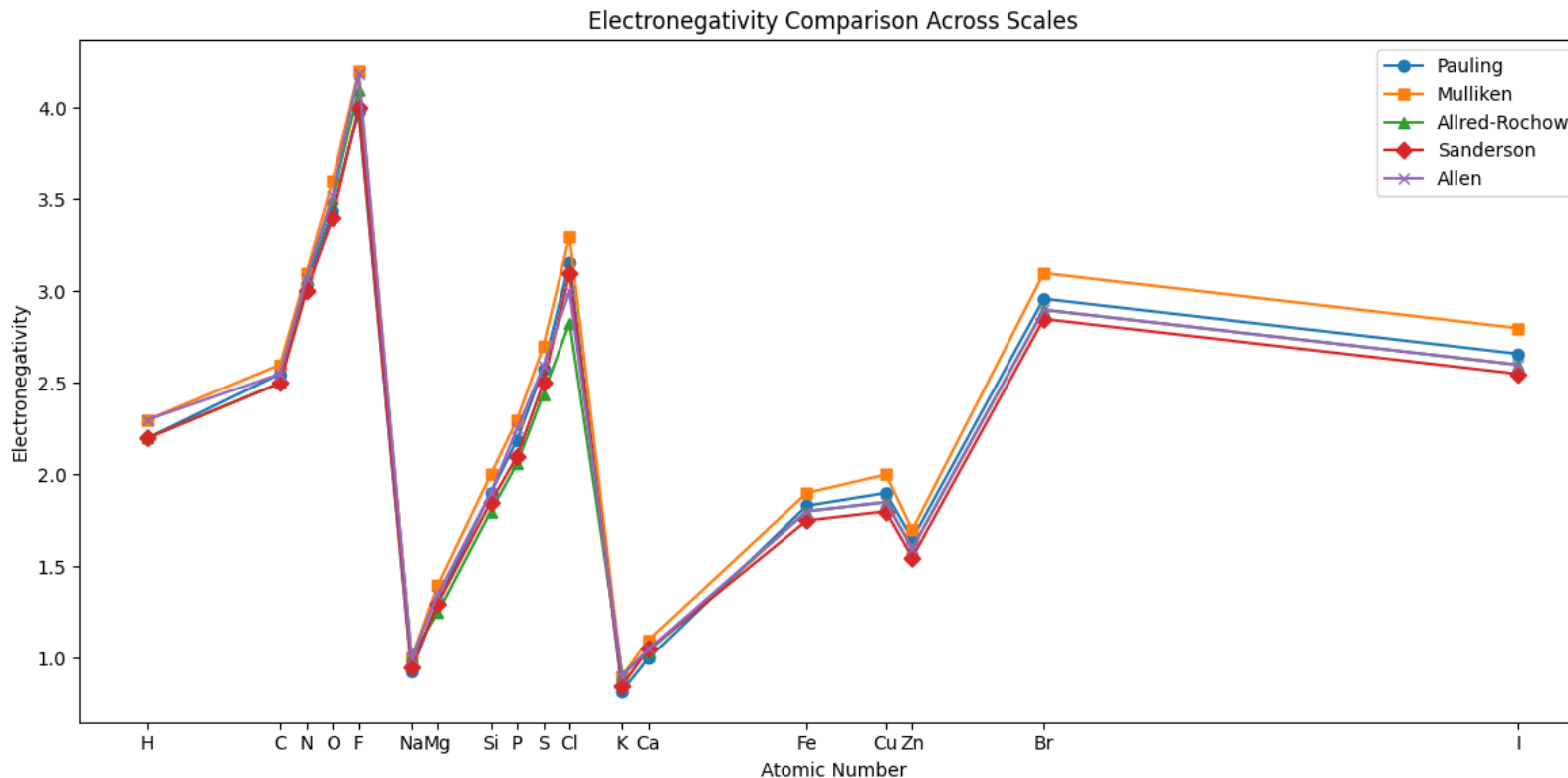


Figure. Electronegativities of the elements. Note the trends across a period and down a group.

Rank the elements carbon (C), nitrogen (N), oxygen (O), and fluorine (F) from least to most electronegative. Enter your ranking as a string of element symbols with no spaces or other delimiters (e.g., “CNOF”).

- **Bond enthalpy (bond dissociation enthalpy)**, a measure of the strength of covalent bonds, is defined as the enthalpy change when 1 mol of bonds is broken **homolytically**.



- Usually reported in kilojoules per mole (kJ/mol) as an average for cleavage of X–Y bonds in many different molecules
- BDEs are always positive because **bond cleavage is endothermic** and **bond formation is exothermic**.
- Stronger bonds have higher BDEs. Triple bonds are stronger than double bonds, which are stronger than single bonds. Stronger bonds are generally shorter than weaker bonds (but not always—see above!).

- Bond enthalpies can be used with a list of bonds made and broken in a reaction to estimate $\Delta H^\circ_{\text{rxn}}$.

reactants $\xrightarrow{\text{cleave all bonds broken}}$ fragments $\xrightarrow{\text{form all bonds made}}$ products

$$\Delta H^\circ_{\text{rxn}} \approx \sum_i^{\text{bonds broken}} \Delta H^\circ_{BD,i} - \sum_j^{\text{bonds made}} \Delta H^\circ_{BD,j}$$

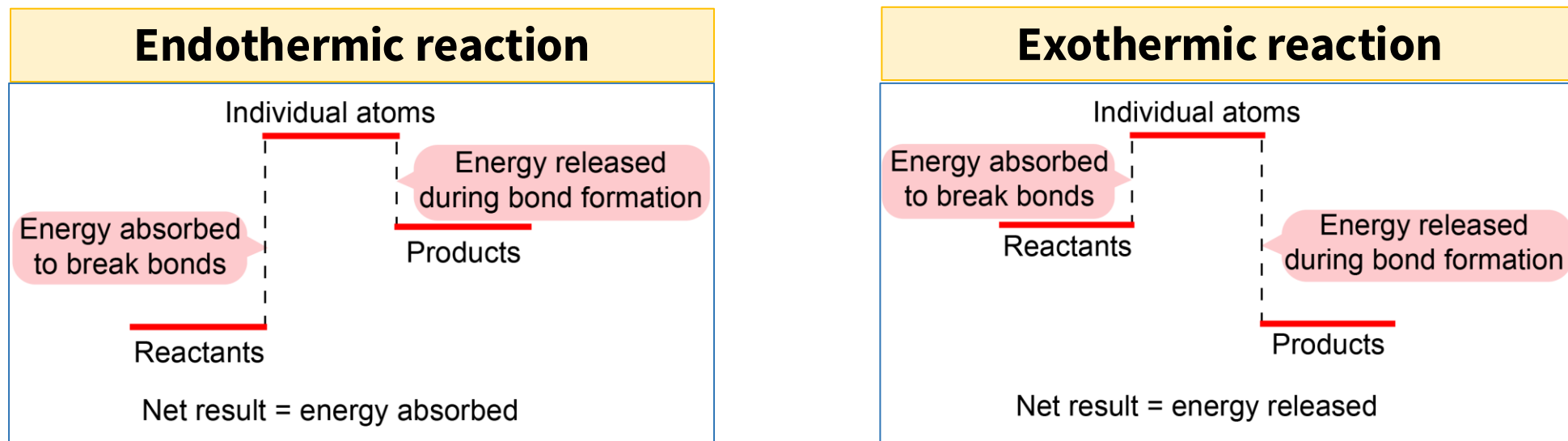


Figure 10.19. Bond enthalpies in endothermic and exothermic reactions.

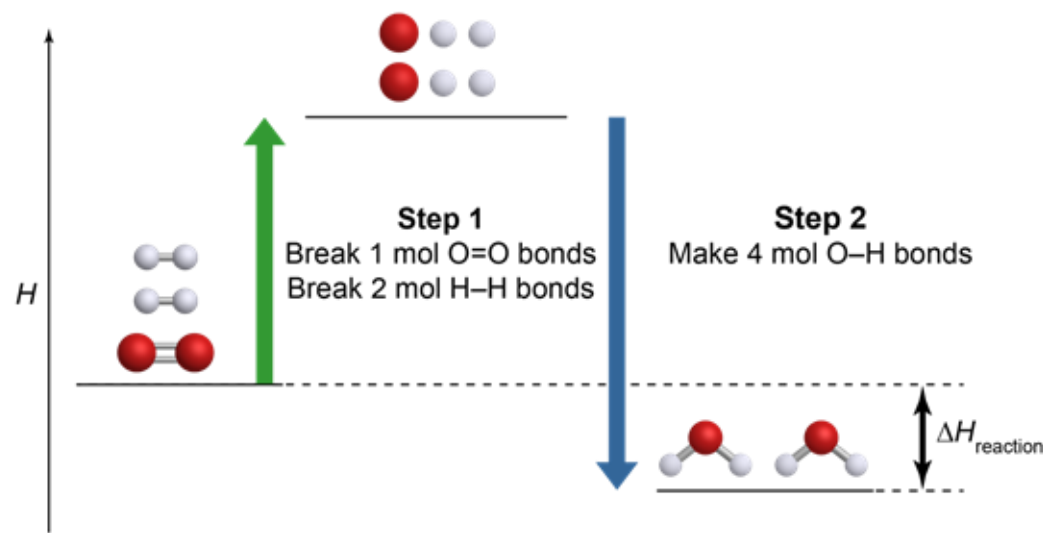
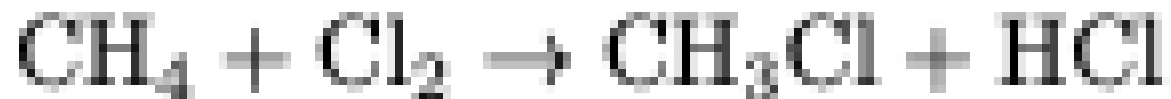


Figure 10.20. Breaking and making bonds in the synthesis of water.

Use bond enthalpies to estimate the enthalpy of reaction for the gas-phase chlorination of methane to give methyl chloride and hydrogen chloride. Relevant bond enthalpies are listed below. Enter your response in kilojoules per mole (kJ/mol) to the nearest 1 kJ/mol with appropriate sign.



C-H: 413 kJ/mol

Cl-Cl: 239 kJ/mol

C-Cl: 339 kJ/mol

H-Cl: 427 kJ/mol

Chapter 10.B

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10.2 Introducing Lewis Structures

- We represent the structures of molecules using element symbols, lines, and dots in a **Lewis structure**. *Only* valence electrons are included in Lewis structures but all valence electrons should be accounted for.
- Bonds may be single, double, or triple (see above).
- **Nonbonding lone pairs** are valence electrons that do not participate in bonding. They are represented as dots near the element symbol to which they “belong.”

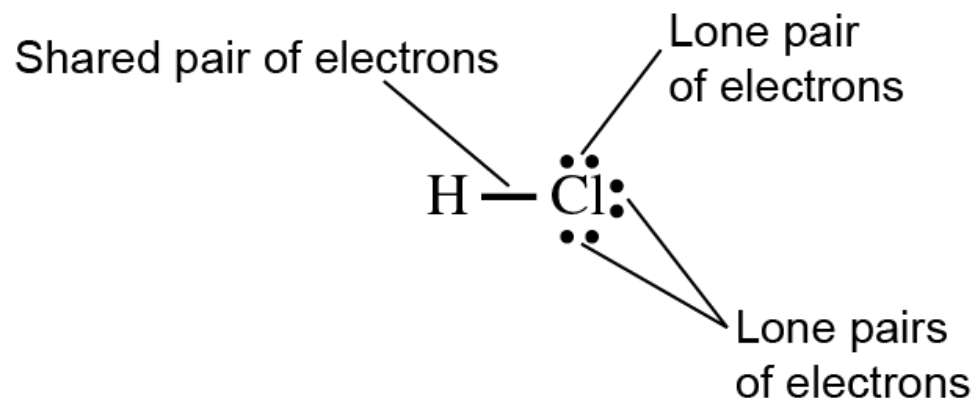


Figure 10.2. A Lewis structure for hydrogen chloride.

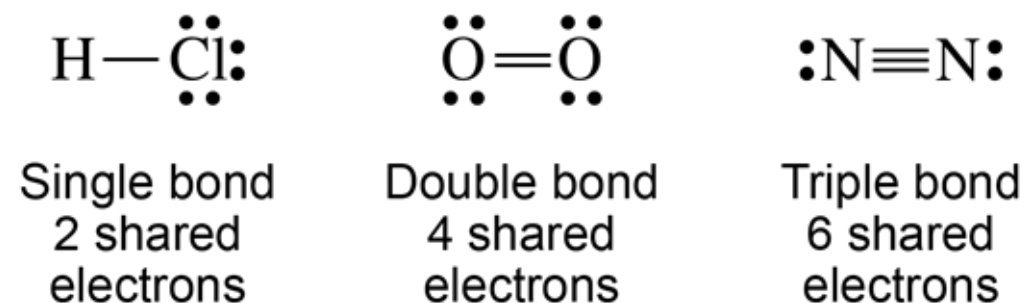


Figure 10.3. Examples of single, double, and triple bonds in Lewis structures.

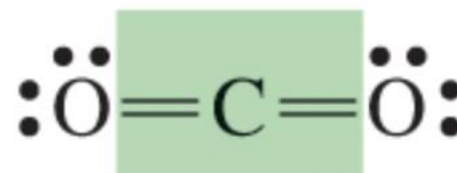
- It has been observed that atoms frequently form enough bonds to achieve a total electron count of eight. This guideline is known as the **octet rule**.
 - The octet rule is **particularly strong** for N, O, and F.
An important exception is hydrogen (H), which always forms just one bond.
 - *Why eight?* For a *p*-block element, this corresponds to a full atomic valence shell (ns^2np^6) and the maximum number of what we'll later call bonding molecular orbitals.

10.2 Counting Electrons in Lewis Structures

- Note that the total number of electrons in a Lewis structure must equal the total number of valence electrons in the atoms of the molecule.
- In the Lewis structure of carbon dioxide, note the total number of electrons and the octet at each atom.



(a)



(b)

Figure 10.4. Lewis structure of carbon dioxide highlighting the octets at oxygen (a) and carbon (b).

10.2 Drawing a Lewis Structure



Drawing a Lewis structure given a molecular formula...

1. Determine the total number of valence electrons. For cations, subtract one electron for each positive charge. For anions, add one electron for each negative charge.
2. Draw a **skeletal structure** of the molecule or ion, arranging the atoms around a central atom. (Generally, the element with smallest Z_{eff} should be placed in the center.)
Connect each atom to the central atom with a single bond (one electron pair).
3. Distribute the remaining electrons as lone pairs on the peripheral atoms (except hydrogen), completing an octet around each atom.
4. Place all remaining electrons on the central atom.
5. Rearrange the electrons of the outer atoms to make multiple bonds with the central atom to obtain octets wherever possible.
6. Assign **formal charges**, making sure no atom has charge greater than ± 1 and the sum of all charges equals the total charge of the molecule or ion.

Example. Draw a Lewis structure for H_2O_2 .

Example. Draw a Lewis structure for CH_3OCH_3 . Think carefully about how the atoms are connected. How can the skeletal structure be inferred from the way the formula is written?

Draw a Lewis structure for SiCl_4 . How many nonbonding lone pairs does the structure contain?
Enter your response as a numeral.

- **Formal charge** is based on the count of valence electrons “belonging” to each atom in the structure.
 - May resemble the more realistic atomic partial charge...
 - ...*but* often does not due to resonance delocalization (more soon).

$$\text{formal charge} = \text{valence electrons in neutral atom} - [\text{nonbonding electrons} + \frac{1}{2}(\text{bonding electrons})]$$

- Over time, you should develop the ability to infer formal charge from common **bonding patterns** of the *p*-block elements—*NOT* by using the formula above. The formula is just an “analytical failsafe” for unfamiliar cases.

Example. Draw a Lewis structure for carbonate anion (CO_3^{2-}).

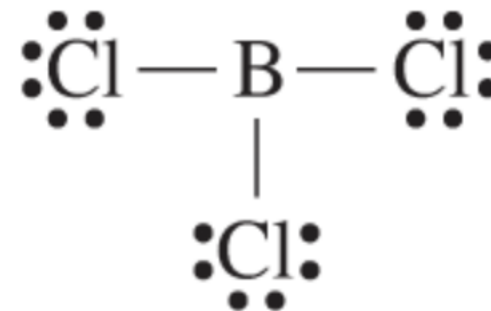
The carboxylate group is CO_2^- . Draw a Lewis structure for HCO_2^- , which includes this group. This structure will include at least one atom with formal charge. Which answer choice best describes the atom(s) with formal charge?

- A. Hydrogen has a formal charge of -1 .
- B. Carbon has a formal charge of -1 .
- C. One oxygen atom has a formal charge of -1 .
- D. Both oxygen atoms have formal charges of -1 .
- E. All atoms in this structure are neutral.

- The octet rule is best applied to second-row elements in groups 14 – 17 (C, N, O, F). There are a few general exceptions to the “rule.”
- **The duet rule.** Hydrogen forms one and only one bond to achieve a “duet” of electrons ($1s^2$).
- **Incomplete octets.** Group 13 elements such as B, Al, and Ga only form three bonds to remain formally neutral. For a similar reason, Be in group 12 forms only two bonds (but Be rarely bonds covalently).



(a)



(b)

Figure 10.7. Exceptions to the octet rule involving hydrogen and beryllium (a) and boron (b).

- The octet rule is best applied to second-row elements in groups 14 – 17 (C, N, O, F). There are a few general exceptions to the “rule.”
- **Expanded octets.** Elements in periods 3 and below can and often do bear more than eight electrons. This is common when an element is linked to more atoms than its period 2 congener suggests it “should” be.

Example. Consider the Lewis structures of H_2SO_4 , H_2S , and H_2O .
Where is the octet rule violated?

10.4 Practice with Formal Charges and the Octet Rule



The nitro group -NO_2 is an important electron-withdrawing group in organic molecules. Draw a valid Lewis structure for CH_3NO_2 , which contains the nitro group. Your structure will include nonzero formal charges; make sure you understand why!

Dimethyl sulfate, $(\text{CH}_3\text{O})_2\text{SO}_2$, was used as an alkylating reagent in the days of yore before it was found to be extremely toxic. Draw a valid Lewis structure for this molecule and indicate where it contains an expanded octet. What is the total electron count at the atom that has an expanded octet?

Hint: to figure out the connectivity, compare to sulfuric acid above.

Chapter 10.C

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Chemical Principles I

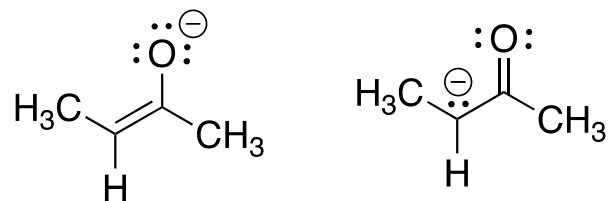
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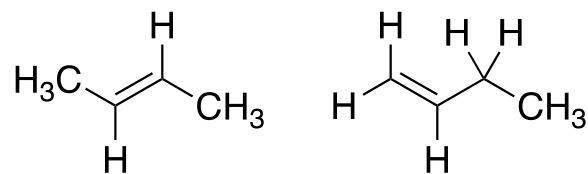
- In Lewis structures, electrons must be located either on an atom or between two atoms. But in many molecules, electrons are **delocalized** over three or more atoms, meaning they cannot be represented with only one Lewis structure!
- Molecules that exhibit electron delocalization are represented using multiple Lewis structures called **resonance forms**.
- Resonance forms differ *only* in the placement of multiple bonds and lone pairs—**nuclei and single bonds do not move.**

Which of the following pairs of structures are *NOT* related as resonance forms? Carbon atoms are located at the intersections of bonds without an atom label.

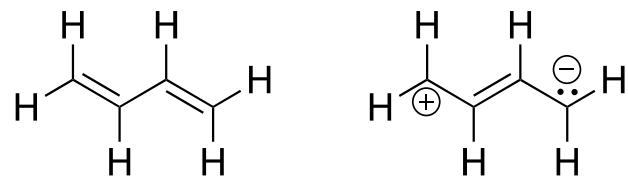
A.



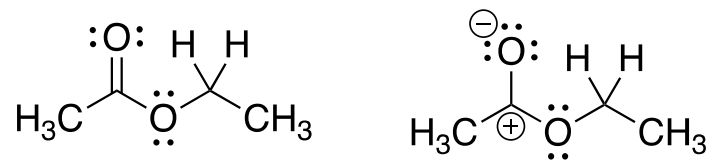
B.



C.



D.



10.3 The Resonance Hybrid; Practice with Resonance Forms



- **Resonance hybrid:** the weighted average of all valid and significant resonance forms for a molecule. Valid structures have reasonable formal charges (-1 , 0 , or $+1$) and “respect the octet guideline.”
 - The resonance hybrid resembles the “true” structure of the molecule.
 - The weights, known as **contributions**, depend on “how important” the structure is to the resonance hybrid.

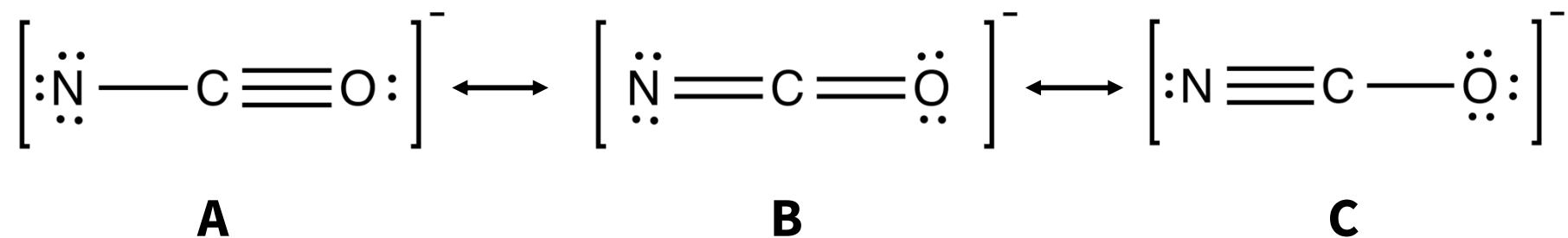
Example. Draw a valid Lewis structure for the nitrite anion (NO_2^-). Then, draw an alternative resonance form.

Let's revisit the carbonate anion (CO_3^{2-}). Draw structures for the three valid resonance contributors for this molecule and the resonance hybrid. Which statement best describes the negative charge in this molecule?

- A. The negative charge is localized on carbon.
- B. The negative charge is localized on a single oxygen atom.
- C. The negative charge is shared unequally by the oxygen atoms.
- D. The negative charge is shared equally by the oxygen atoms.

- *How do we rank the relative contributions of resonance forms?* There are three ideas to consider.
 1. Fewer and smaller formal charges are preferable to larger charges.
 2. Avoid like charges (++) or (--) on adjacent atoms and separate like charges as much as possible. Keep opposing charges (+-) as close as possible.
 3. **Negative** formal charge should be located on **more** electronegative atoms.
Positive formal charge should be located on **less** electronegative atoms.

Assign formal charges to all the atoms in the three resonance forms of NCO^- labeled **A** – **C** below. Rank the structures from greatest (1) to smallest (3) contributions to the resonance hybrid. Enter your ranking as a string of letters with no delimiters between them (e.g., “ABC”).



10.5 Polar Covalent Bonds

- When two atoms of different electronegativity engage in a covalent bond, one atom has greater electron density than the other.
- One end of the bond has partial positive charge and the other partial negative charge—a **dipole**.
- **Polar covalent bonds** have substantial dipoles.

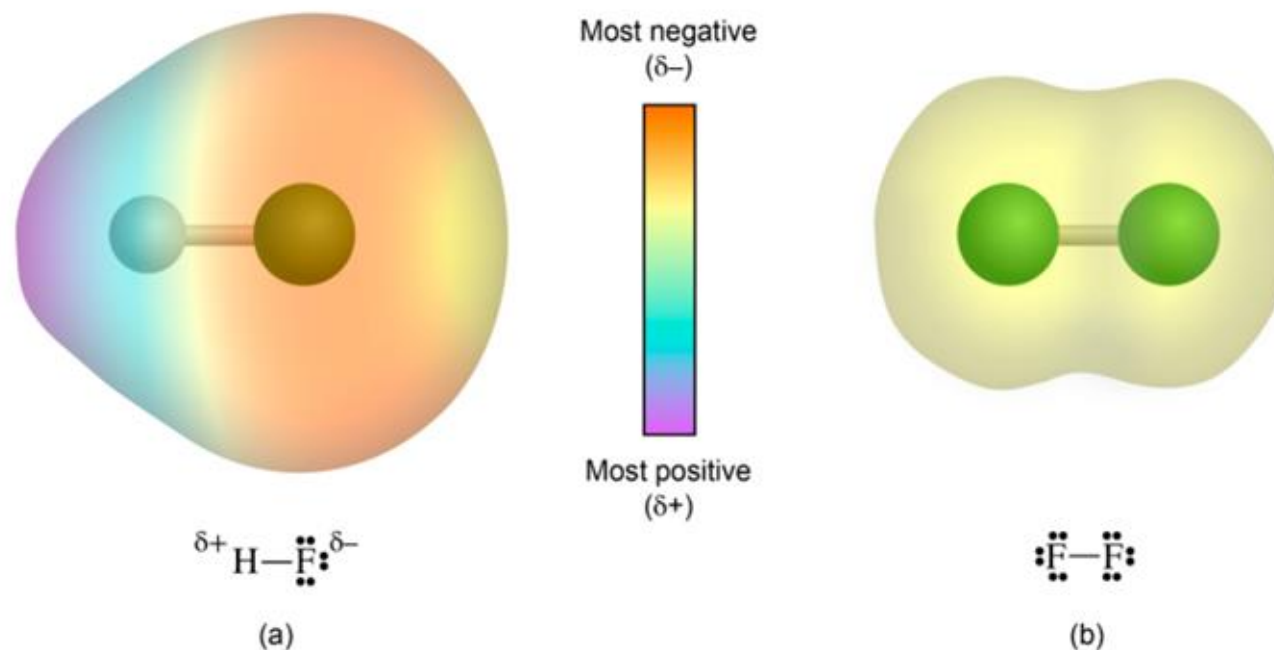


Figure 10.16. (a) Unequal electron sharing in HF and (b) equal electron sharing in F_2 .

10.5 Bond Polarity is a Continuum

- The difference in electronegativity between two atoms in a bond (ΔEN) can be used to classify bonds on a continuum from fully covalent to fully ionic.

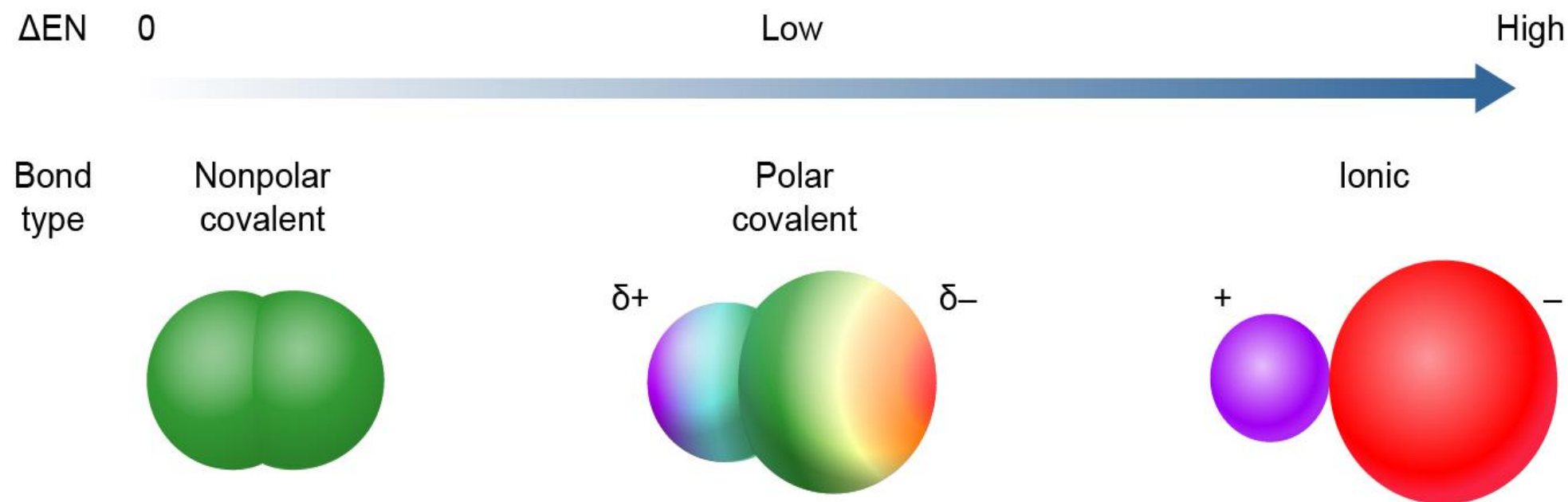


Figure 10.15. The bonding continuum.

- Some of the amazing things we can now do with covalent bonding...
 1. Use graphs of potential energy as a function of internuclear distance to evaluate the lengths and strengths of covalent bonds.
 2. Define and apply the concept of electronegativity; apply periodic trends in electronegativity to rank a series of elements.
 3. Estimate enthalpy of reaction using bond enthalpies and a list of bonds made and broken.
 4. Draw Lewis structures for simple small molecules and ions given a molecular formula and, where necessary, information about connectivity.
 5. Draw and interpret resonance forms for molecules exhibiting electron delocalization.
 6. Combine resonance forms to generate the resonance hybrid, an average representation of molecular structure.
 7. Apply differences in electronegativity to classify a covalent bond as polar or nonpolar and draw bond dipole vectors.